

High-Level Design (HLD): CSVAT

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1 Subsystem Design

1.1 Frontend Subsystem (React)

- **BoundarySelector:** Component for searching Tehsils/Villages or handling Drag-and-Drop GeoJSON.
- **ReportViewer:** The container for the "Data Story." It dynamically renders sections based on the analytics result JSON.
 - **Section 1:** Interactive Map (Layer overlays).
 - **Section 2:** Charts (Time-series trends).
 - **Section 3:** Narrative Text (Generated insights).
- **ExportManager:** Handles PDF generation and JSON-to-CSV conversion for raw data download.

1.2 Report Subsystem

Unlike a static file server, the "Report" is a dynamic view state.

- **Interactive HTML:** Interactive report is rendered in the React UI as a structured "Data Story".
- **PDF Generation:** PDF export is supported in both execution modes. The specific generation strategy (client-side export vs backend export) is TBD.
- **CSV Export:** Supported via direct conversion of the results JSON.
- **Failure Handling:** On failure, the system returns an explicit error message and does not publish partial analytics outputs.

1.3 Data Persistence (High-Level)

- Store minimal job metadata (id, params, status, timestamps).
- **Job Status Model:** Standardized states are PENDING, RUNNING, SUCCESS, FAILED.
- **Intermediate Persistence:** Intermediate rasters and temporary clipped layers are not persisted beyond job execution (per SRS). Only final result JSON and exported CSV/PDF are stored.
- Optionally cache resolved boundaries to improve performance.
- **Storage technology:** PostGIS (as per SRS).

2 Proposed API Contract (REST)

The Backend exposes a RESTful API over HTTPS.

Authentication: Bearer Token (JWT).

2.1 Core Endpoints (Proposed)

Endpoint	Purpose	Requirement
GET /api/v1/boundaries/search?q={name}	Search for admin boundaries.	FR-BA-02
POST /api/v1/boundaries/validate	Upload GeoJSON to check intersection with active Tehsils.	FR-BA-03
GET /api/v1/layers	List available analytics layers and year ranges.	FR-CI-01
POST /api/v1/jobs	Submit a new analysis job (supports mode selection).	FR-CI/SW
POST /api/v1/jobs/save	Save client-side run metadata/results (WASM Mode only).	Data Persistence
GET /api/v1/jobs/{id}	Get status and final results (JSON).	FR-CI-04
GET /api/v1/jobs/{id}/assets/{type}	Get specific asset URL (e.g., CSV download).	Outputs / Reporting

2.2 Job Payload Example

Request (POST /api/v1/jobs):

```
{
  "boundary_id": "vg-12345",
  "layers": ["cropping_intensity", "surface_water"],
  "years": [2020, 2021, 2022],
  "mode": "SERVER"
}
```

Response (GET /api/v1/jobs/{id} - Success):

```
{
  "id": "job-abc-999",
  "status": "SUCCESS",
  "results": {
    "cropping_intensity": {
      "2020": { "single": 45.2, "double": 10.5 },
      "2021": { "single": 40.0, "double": 15.2 }
    }
  }
}
```

3 Detailed Analytics Pipelines

3.1 Cropping Intensity (FR-CI)

Input: Village Boundary + Year Range.

Fetch: LULC Raster stack from CORE Stack.

Process:

1. Mask raster by boundary.
2. Remap pixel values (1=Single, 2=Double, 3=Triple).
3. Count pixels per category.
4. Convert pixel counts to Hectares (using pixel resolution).

Output: Timeseries JSON of areas per class.

3.2 Seasonal Surface Water (FR-SW)

Fetch: Surface water datasets retrieved from CORE Stack (exact source layers TBD).

Logic:

- **Perennial:** Water present > 9 months.
- **Seasonal:** Water present 1-8 months.

Output: Bar chart data comparing Seasonal vs. Perennial area.

3.3 Vegetation and Degradation Tracking (FR-VD)

Input: Village Boundary + Time Period.

Fetch: Vegetation/LULC rasters from CORE Stack.

Logic:

- Compare vegetation class extent between years.
- Identify pixels transitioning from Vegetation to Non-Vegetation (Loss).
- Estimate degraded land area based on vegetation-to-non-vegetation transitions and relevant degradation classes defined in CORE Stack LULC taxonomy.

Output: Tabular summary of gain/loss and degradation estimates.

4 Traceability Matrix

Component	Requirement ID	Description
BoundarySelector (UI)	FR-BA-01, FR-BA-02	Handling User Inputs & GeoJSON uploads.

FastAPI Validation	FR-BA-03, FR-BA-04	Validating boundary intersection logic.
Analytics Engine (Shared Logic)	FR-CI, FR-SW, FR-VD	Core algorithms for cropping, water, and vegetation.
Backend Execution Adapter	FR (All) - Server Mode	Orchestrates Celery/GEE execution.
WASM Execution Adapter	FR (All) - Client Mode	Orchestrates in-browser WebAssembly execution.
ExportManager	FR-VD-04	Generating PDF/CSV outputs.

5 Out of Scope

- Full historical job dashboard (UI is session-focused).
- Multi-village batch comparisons or district-level aggregations.
- User management features beyond authentication (e.g., profile editing, role management).
- Long-term storage of intermediate raster files (only final JSON statistics are persisted).